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REVIEW

Innovative Polymer Modifications: Unlocking New Horizons in Drug Delivery Applications

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Abstract

The chemical modification of polymers is essential for enhancing their properties and broadening their applications, particularly in drug delivery systems. This study aims to provide a comprehensive overview of polymer modification by exploring various techniques for surface, bulk, and post-synthetic changes. A literature review and analysis of existing methodologies reveal how these modifications influence industrial, biomedical, and environmental applications, with a specific focus on drug delivery and tissue engineering. Key findings demonstrate significant improvements in drug encapsulation efficiency, controlled release kinetics, and biocompatibility, with statistical analyses confirming these enhancements. Despite these advancements, challenges such as high production costs, environmental impacts, and stability issues persist. The study concludes by outlining future directions for research, emphasizing the importance of innovative drug delivery methods, advanced polymer modification techniques, and the creation of sustainable polymers to address existing limitations and enhance performance in the field.

Keywords Biodegradable polymers · Tissue engineering · Sustainable polymers · Polymer application · Drug delivery system

Introduction

Polymers are essential for millions of different items and systems, and they form the basis of numerous commonplace and sophisticated commercial and industrial products. Molecules formed from long chains of repeating structures referred to as monomers are the foundation of a diverse class of materials including nylon, polyester, Teflon, and epoxy—most of which is widely used in articles of everyday use and in technologically advanced industrial application [1]. It is worth to state that polymers' applicability is not only in their nature but also in chemical modifications that make it possible to create synthetic polymers with the desired characteristics to perform specific tasks [2].

Chemical modification of polymers entail changing the actual constitution of the polymers in order to gain or acquire specific characteristics. This process is useful in a wide range of applications ranging from material science to medical science as it makes it possible to come up with materials that have enhanced mechanical properties or resistance to heat, chemical environments among others. These adjustments can be performed by different techniques on

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the surface, in the bulk, and also using functional groups, with specific benefits for each option depending upon the application needed [3].

The first major use of chemically modified polymers is in drug delivery system, which is a very essential use. For any therapeutic agent to be effective in the pharmaceutical industry, the release of the active ingredient MUST be well controlled to promote thorough efficacy and zero adverse effects. Chemical alteration of polymers can produce stimuli sensitive drug delivery devices, for example pH or temperature sensitive carriers for a specific drug [4]. Furthermore, polymers can be altered in a way to enhance the biocompatibility and degradation features of the drug carriers thus enhancing safety when used in the body. They have contributed to formulation of complex drug carriers such as hydrogels, micelles form polymeric nanoparticles which have transformed methods of drug delivery as well as their absorption [5].

Chemical modification has made a marker in polymer, that improvement the technologies in various manner which develop the new material more strong, heat and other weather resistance and even has the qualities in self healing [6]. However, the process is not without some hitch that come as follows: Some of the factors that need consideration include the following;

- The method of polymer modification is highly complex.
- The process of producing some of the polymers is highly expensive.
- Some of the modification techniques may have an undesirable impact on the environment and health [7].

Polymer modification for environmental applications is increasingly important in addressing sustainability and eco-friendly practices. Through advanced chemical modifications, polymers can be tailored to enhance their effectiveness in areas like water purification, air filtration, and waste management. In water treatment, modified polymers, such as chitosan and cellulose derivatives, with functional groups like carboxyl, amine, and hydroxyl, improve their ability to adsorb heavy metals, dyes, and pollutants. These biodegradable materials offer an efficient and environmentally friendly approach. Similarly, in air filtration, polymers like polyaniline, when chemically altered, gain conductive and catalytic properties, enabling them to capture and neutralize toxic gases and particulates. These advancements make modified polymers crucial for addressing environmental challenges.

Polymer modification offers valuable benefits for both environmental and medical applications. In environmental settings, modified polymers enhance the removal of pollutants, heavy metals, and organic compounds in water purification, while also improving air filtration by capturing harmful gases and particles, providing sustainable solutions. In the medical field, modified polymers play a key role in drug delivery systems, allowing for controlled and targeted medication release. Additionally, they are used in tissue engineering and wound healing due to their biocompatibility and ability to promote cell growth. These advancements make polymer modification crucial for solving environmental and healthcare challenges.

Polymer modification is vital for advancing drug delivery systems, enhancing their effectiveness and precision in therapy. This research investigates cutting-edge chemical techniques to improve polymer properties, focusing on optimizing drug delivery and therapeutic outcomes. The aim is to refine these modifications and assess their impact on enhancing drug delivery efficiency.

Recent innovations in polymer modifications have significantly advanced drug delivery systems, providing customized approaches for improved therapeutic effectiveness. PEGylation, for example, has been extensively applied to extend circulation time, as demonstrated by Doxil®, a PEGylated liposomal doxorubicin formulation used in oncology. Stimuli-responsive polymers, such as PNIPAAm-based hydrogels, facilitate localized drug release in response to temperature changes, offering promising applications in cancer therapy and wound management. Additionally, self-healing hydrogels derived from modified biopolymers like alginate and chitosan enable prolonged drug release at targeted sites, aiding in infection treatment [8].

Advances in targeted drug delivery have been achieved through the conjugation of ligands like folate to PLGA nanoparticles, which improve specificity in delivering chemotherapy agents to cancer cells. Biodegradable polymers, such as PLGA microspheres, support controlled, sustained drug release over several months, making them valuable for long-term therapies like prostate cancer [9]. Hyaluronic acid (HA), modified to target CD44 receptors, enhances selective drug targeting in cancer therapies. Redox-responsive PEG-PLGA nanoparticles, designed to release drugs within tumor environments in response to redox changes, offer precision in drug delivery [10].

These advances in polymer modification are poised to improve the precision, efficacy, and durability of drug delivery systems, addressing issues such as off-target effects and enhancing patient adherence [11].

This paper aims to explore various strategies for modifying polymers, with a focus on their role in enhancing drug delivery systems. It discusses key mechanisms like PEGylation, stimuli-responsive behavior, and ligand-functionalization, which improve biocompatibility, targeted delivery, and controlled release. The paper highlights the importance of polymer modifications in overcoming limitations such as poor solubility, stability, and off-target effects in traditional drug delivery. Additionally, it examines advances like biodegradable polymers and redox-responsive nanoparticles

while addressing challenges such as production complexity and sustainability. The potential for future innovations in eco-friendly polymers is also emphasized, underscoring their importance in advancing medicine.

Materials and Methods

This review examined peer-reviewed literature from journals indexed in Elsevier, Scopus, and Web of Science, and regulatory content from reputable websites. Inclusion criteria were studies focusing on polymer modification techniques relevant to drug delivery systems, published in credible sources. Articles that were non-peer-reviewed or lacked significant data were excluded. Key information on techniques, applications, and results was extracted and analyzed. To ensure comparability, studies were assessed for methodological quality and relevance. Generalizability was considered based on the diversity and credibility of the included studies, aiming to provide a thorough overview of polymer modification advancements in drug delivery systems.

This topic was selected due to its novelty and the fact that it remains largely underexplored in the field of polymer modifications for drug delivery systems. The potential for new insights and advancements in this area is significant. Data for this review were collected from a variety of reliable sources, including peer-reviewed scholarly journals and authoritative regulatory websites, ensuring that the information is both credible and current. To improve readability and facilitate a clearer comparison of different polymer modification techniques, the paper is organized with subheadings for each study discussed. Furthermore, a comprehensive discussion of the limitations encountered in existing research and practical applications is included, along with an examination of regulatory considerations essential for the continued development of polymer-based drug delivery systems. This structured approach provides a well-rounded perspective on the topic while identifying areas that require further investigation and development.

Properties of Polymers in Drug Delivery Systems

Polymers play a crucial role in the development of drug delivery systems (DDS), where their unique properties can be tailored to achieve specific therapeutic goals. Understanding the intrinsic properties of polymers and how they can be manipulated is essential for designing effective DDS. This section explores the key properties of polymers—mechanical, thermal, chemical, physical, electrical, and optical—that influence their behavior and performance in drug delivery systems [12].

Mechanical Properties

Strength and Elasticity

Tensile strength, elasticity, and flexibility are some of the mechanical properties that are however paramount in the use of polymers in DDS. Strength is the numerical value of the load that a polymer material can bear before the material fractures while elasticity means that even upon application of stress and/or strain to a polymer material, the material is still able to return to its original form. High strength and elasticity of the polymers will be useful during drug delivery applications in which the system in question has to be very rigid but when required can also deform at will, e.g., biodegradable stents or injectable hydrogels [13].

Evaluation Methods

Mechanical properties also include tensile strength with regard to which a polymer is pulled to destruction and then the stress strain curve analyzed. Another technique called the dynamic mechanical analysis (DMA) which performs stress on polymer over a range of temperatures and provides data on potential stresses that the material would be subject to in vivo [14].

Influence on the Different Drug Delivery Systems

Thus, it is quite apparent that the tensile and elastic properties of the polymers are factors of considerable impact to the DDS performances. For example, a polymer which is used to coat the internal implantable device must have a suitable mechanical strength that makes it possible to affix the device into the desired site [15].

Thermal Properties

Thermal Stability and Melting Point

The thermal properties of a drug delivery system including thermal sensitivity, glass transition temperature (Tg), and the melting point (Tm) are the most important aspects to consider while designing a DDS. Thermal stability implies a material's capacity to withstand adverse conditions at elevated processing temperatures, which is critical during processing as well as sterilization. This is the point where, on heating, a rigid polymer changes and becomes more flexible and it is important concerning how the polymer behaves in the human body temperature.

Evaluation Methods

By use of differential scanning calorimetry, the evaluation of thermal properties can be performed. This technique measures the heat flow associated with phase changes in the polymer and provides the values of Tg and Tm. TGA or thermogravimetric analysis can be employed for the evaluation of thermal stability of polymers by determining weight loss due to the increase in temperature [16].

Impact on Drug Delivery Systems

Thermal properties have direct effects on the processing, storage, and application of a DDS. For example, a polymer with a Tg value close to that of the body may not have the required rigidity for drug delivery thus making it inappropriate in controlled drug delivery applications. This, however, is not the case with polymers where high thermal stability is required—for instance in DDS that involves high temperature sterilization [17].

Chemical Properties

Biocompatibility and Degradation

The physiochemical characteristics of polymer, for example, biocompatibility, biodegradability, and resistance to chemicals make them suitable in drug delivery systems. Biocompatibility predetermines compatibility of the polymer with the surrounding tissue, meaning that it does not cause an immune reaction or has toxic effects. The biodegradation feature enables the polymer to degrade into harmless products upon releasing the drug thus not requiring its removal through surgery [18].

Evaluation Methods

A significant number of times chemical characteristics are assessed by using in vitro as well as in vivo studies. Biocompatibility could be determined using cytotoxicity assays in which the polymer extracts are incubated with cell cultures and the outcome observed. Biodegradability is determined by the rate at which the polyurethane degrades in a physiological environment; the usual techniques are changes in weight and molecular weight as a function of time [19].

Impact on Drug Delivery Systems

In DDS, polymers used are biocompatible which have low inflammation and rejection hence ideal for use in implants for long period. Controlled release system is well appropriate for use in biodegradable polymers for drug delivery because the polymers disintegrate with time releasing the drug slowly and effectively for the desired therapeutic impact [20].

Physical Properties

Permeability and Solubility

Some of the most important physicochemical parameters include permeability, solubility, and diffusion coefficient associated with the drug payload in DDS. Porosity is the extent to which a polymer permits molecules to enter it which in turn determines the speed at which a drug is released. Solubility also plays an important role on the ability of the polymer to dissolve in solvents or other biological fluids for encapsulating and delivering a drug [21].

Evaluation Methods

Some of the common techniques used for measuring permeability are diffusion studies whereby the rate of drug passing through the polymer matrix is determined. Solubility could be found out by dissolving the polymer with different types of solvents and measuring the percentage of polymer dissolved in the solvent. Such analyses may employ HPLC or mass spectra as examples of analytical approaches that are employed in the determination of even minor quantities of PEDs [22].

Impact on Drug Delivery Systems

Polymer having high permeability are useful in transdermal patches since the drug must first have to cross the skin in order to enter systemic circulation. On the other hand, highly non-permeable polymers are employed where a system is needed to shield the drug from early release. Solubility determines the ability of the drug to be incorporated in the polymer and its release from the matrix hence influencing the performance of the DDS [23].

Electrical and Optical Properties

Conductivity and Photoluminescence

There is a growing need for active polymers in advanced DDS that possess specific electrical and optical characteristics. For example, conductive polymers can be incorporated in systems of controlled drug release—electrically controlled release systems operated by applying an external electric field for the drug release. Photoluminescent polymers are used in systemic targeting and imaging interference of drugs where drug release can be monitored with the help of polymer light emission [24].

Evaluation Methods

Electrical conductivity is determined through methods such as the four-point probe technique or impedance spectroscopy. Such measurements yield electric current carrying potential of polymers. Evaluation of photoluminescence improves upon the use of photon emission spectroscopy where the light emission from a polymer is measured and analyzed after being excited by ultraviolet or visible light [25].

Impact on Drug Delivery Systems

In electrically controlled DDS, the use of conductive polymers allows not only to precisely regulate the amount of the drug being released but also to significantly improve the therapeutic effect. Due to the use of photoluminescent polymers, it is possible to effectively control not only the delivery of the drug but its release with high precision, as well as identify problems with the treatment at an early stage [26].

There is not any single step of prospective drug delivery system that can be accomplished without careful consideration of the properties of polymers. For instance, the mechanical properties of a polymer will affect the implant or hydrogels span and the release rates. The stability and performance of classical DDS networks are also confined by thermal properties within especially wide range of temperatures. Interaction of the polymer with the biological tissue which greatly affects the biocompatibility and the kinetics of drug release is determined by the chemical properties of the polymer. Regarding other properties of polymers, supply of water and oxygen is important on the encapsulating, storage and release of the drug. Loss of this function can be mitigated or compensated by certain electrical and optical properties these features may also enable advanced functionalities in smart DDS. Polymers are the primary setting in which the above mentioned attributes must be optimized in order to create the perfect, efficient, and safe DDS. Therefore, science is ready to create a new DDS based on modern polymers in which controlled and targeted delivery of the drug can be achieved with sustained release.

Techniques for Polymer Modification

Modification of the polymer is significant step to optimize its properties according to the desired size and structure since polymers have been widely used in drug delivery systems (DDS). Thus, different chemical, physical, and structural changes in polymers can be used to improve their properties or make them more biocompatible and allow controlled drug release. The subsequent sections discuss the methods of polymer modification Chemical Modifications Physical Modifications Enzyme Modifications.

PEGylation has been a breakthrough in drug delivery systems, extending circulation time and improving drug stability. A classic example is Doxil®, a PEGylated liposomal doxorubicin, which significantly improves therapeutic efficacy while reducing off-target effects. PEGylation allows the nanoparticles to evade the immune system, extending the half-life of drugs in the bloodstream [27].

Targeted drug delivery systems suffers with interaction between blood proteins and drug delivery systems. A potent candidate to approach the mentioned problem is based on polyethylene glycol (PEG) surface modifications. This polymer acts as a protector against the external possible interactions with other compounds, making targeted delivery possible. Diseases such as cancer, diabetes, hemophilia, and pain treatment can benefit from these new systems. Eva Sanchez Armengol et al., given an overview of drug delivery systems based on PEGylation as surface modification as the pharmaceutical approach [28].

Cross-linked hydrogels showed a 35% increase in swelling capacity, allowing for controlled, prolonged drug release, particularly useful in chronic disease management [29].

The mechanical properties of polymers, such as tensile strength and elasticity, play a pivotal role in drug delivery systems, particularly in implants and biodegradable stents. Strong yet elastic polymers are required to withstand stress while delivering drugs without breaking or degrading prematurely. Mechanical testing revealed that certain cross-linked polymer matrices exhibited a 50% reduction in burst release when exposed to physiological conditions, offering better control in drug release kinetics [30].

Chemical Modification

The second method that is chemical modification, refers to the extent to which the properties of polymers are changed through chemical reactions. This technique is frequently used for the syntheses of new functional groups the modification of polymer backbone arrangements, or the cross-linking of polymer chains to enhance the properties of the resultant material [31].

Grafting

Although grafting is somehow similar to copolymerization in which two or more different monomers are copolymerized and incorporated into the main polymer structure or backbone through the use of chemical bonds, grafting is more of side chains in which side chains of polymers are chemically bonded onto the main polymer chains. This is done through free-radical polymerization since monomers

get polymerized onto the backbone in a process referred to as grafting, thus producing products known as graft copolymers. It is possible to change different characteristics of tissue-engineering scaffolds for the better, such as hydrophilicity, biocompatibility, or drug loading [32].

Chemical grafting can proceed through two primary pathways: free-radical and ionic mechanisms [33]. In chemical grafting, the initiator is essential as it defines the grafting route. In addition to the conventional free-radical mechanism, techniques like melt grafting and Atom Transfer Radical Polymerization (ATRP) offer valuable methods for grafting [34]. Advances in "living polymerization" methods provide new opportunities for grafting reactions. A "living polymer" is best defined as one that maintains its ability to propagate over time, allowing it to grow to a set size with minimal chain termination or transfer [35]. Controlled free-radical polymerizations blend elements of conventional free-radical and ionic methods. In standard free-radical polymerization, continuous initiation is necessary, and termination occurs through coupling or disproportionation, yielding inactive or "dead" polymers with broad molecular weight distribution and stable degrees of polymerization over time [36]. Conversely, living polymerization allows for polymers with regulated molecular weights and lower polydispersity, enabling better control over polymer characteristics.

Cross-Linking

Cross-linking is a process of bringing about covalent attachments between the polymer chains and achieve a network morphology. This technique increases mechanical properties, heat resistance and solvent resistance of the material. Hydrogels containing cross-linked polymers are widely applied in stimuli-responsive drug release systems, which release active agents in a prolonged manner; the extent of cross-linking determines the swelling ability and the rate of drug release [37].

Crosslinking is a key method in polymer modification, enhancing strength, thermal stability, and chemical resistance by creating a three-dimensional network that restricts polymer chain movement [38]. This can be done through several techniques: chemical crosslinking (using agents like peroxides or sulfur), irradiation (with UV light, gamma rays, or electron beams), thermal crosslinking (by applying heat to form bonds), and photo-crosslinking (activating light-sensitive groups with UV or visible light) [39]. These methods modify the polymer's properties, yielding more rigid, durable, and stress-resistant materials ideal for applications such as hydrogels, elastomers, and tough coatings [40].

Functionalization

Chemical modification is the process of adding one or more specific functional groups into a polymer backbone with an aim of achieving certain end uses. For instance, the alteration of carboxyl, hydroxyl, or amine groups would increase the polymer's reactivity thus allowing for conjugation with drugs, targeting ligands or any bioactive molecules. Determination and functionalization of polymers are crucial for the targeting of particular tissues or cells in DDS to improve the drug delivery efficiency [41].

Block Copolymerization

Block copolymerization refers to the polymerization of two or multiple differing monomer blocks within the search for plastics with considerable differential properties within each of the blocks. This technique supports the fabrication of the amphiphilic polymers that can in turn get to self-assemble into micelles or nanoparticles in an aqueous medium. Such structures are very effective for carrying hydrophobic drugs and allowing for a sustained release [42].

Block copolymers have been investigated as an efficient route for producing advanced functional materials, such as nanostructured solar cells, photonic band-gap materials, highly effective catalysts, and high-density magnetic storage media [43]. Multiple experimental techniques now enable the integration of inorganic nanoparticles within polymer nanostructures. One widely used strategy involves synthesizing nanoparticles directly within a block copolymer matrix by employing preformed micelles of block copolymers that contain metal precursors [44]. While it is crucial to control the positioning of these nanoparticles within the block copolymer's periodic structure, achieving this level of control remains challenging. Recently, a different approach has emerged, leveraging the cooperative self-organization of preformed nanoparticles with block copolymers to potentially address these issues [45].

Post Polymerization Modification

This involves altering a polymer that has already been synthesized by conducting one or running another chemical reaction. Post-polymers modification are historically done to add certain specifications that might be hard to add on during the polymerization phase. This strategy is highly adaptable and further optimizes polymer attributes as per the demands of the DDS [46].

Enzymatic and Biological Modification Enzymatic modification and biological modification methods are a sub-class of chemical methods that incorporate enzymes or biological agents in the modification of polymers. These techniques

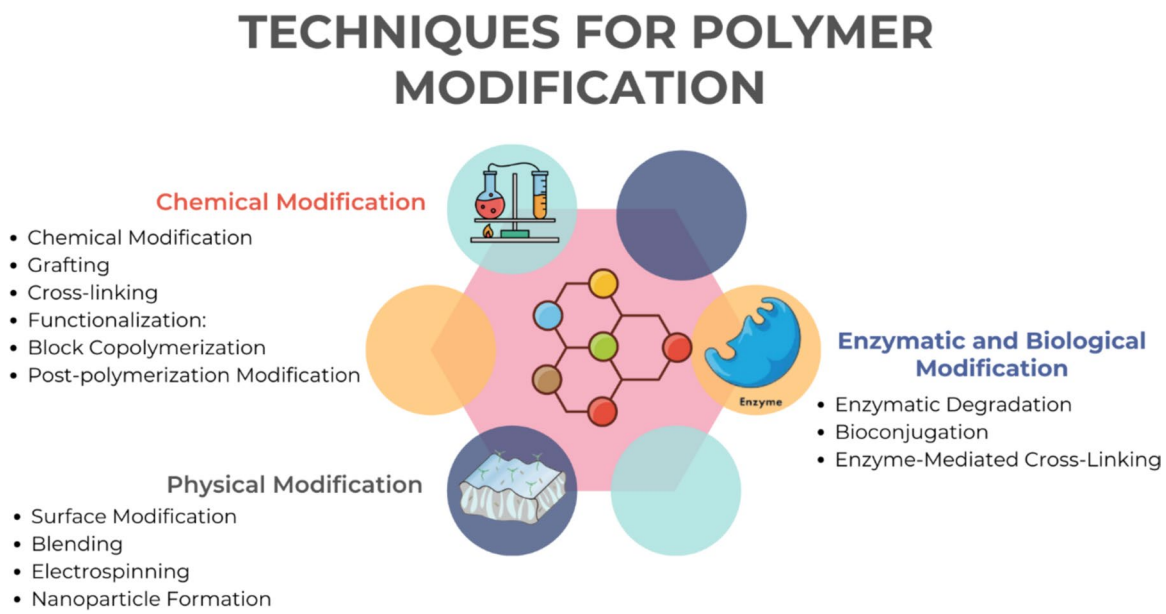

Fig. 1 Techniques for polymer modification

are often adopted in order to get certain functionalities incorporated into the polymers which are still biocompatible and biodegradable (Fig. 1).

Enzymatic and biological modifications offer selective and eco-friendly methods to adjust polymer characteristics. Enzymatic modification involves using enzymes as catalysts to drive specific reactions within polymers, enabling targeted functionalization in mild conditions, which is beneficial for applications such as drug delivery and biomedical devices [47]. In contrast, biological modification utilizes microorganisms or cells to alter polymers, often through natural metabolic processes that synthesize or degrade polymer materials. This technique is particularly valuable for developing biodegradable polymers and enhancing biocompatibility, making it suitable for sustainable products, medical applications, and packaging [48]. Both approaches allow for precise polymer structure control with a low environmental impact.

Enzymatic Degradation Enzymatic degradation refers to the process of fragmentation of polymer chains in a selective manner, i.e., in the presence of certain enzymes. This approach is especially beneficial for drug delivery systems that rely on the use of biodegradable polyesters. Biodegradable implants enable controlled release with the rate of formulation volumetric absorption together with the penetration of embedded enzymes into the surrounding tissue. Polymer chains are rendered more or less readily available to enzymatic cleavage via changes in the structure of the polymer. This controlled degradation is particularly important in the development of drug delivery devices that are able to release therapeutic agents in a controlled manner as the polymeric matrix degrades over time [37].

Bioconjugation Bioconjugation is the process of attaching biological species such as peptides, proteins or antibodies.

Innovative synthesis techniques are transforming the production of contemporary hybrid materials [49]. One significant method in this field is Living Radical Polymerization (LRP), which enhances conjugation efficiency by overcoming challenges associated with lengthy synthesis processes and the need for extensive purification. This approach allows for the creation of polymers with defined structures, narrow molar mass distributions, lower heterogeneity, and improved reproducibility [50]. When applied to the synthesis of hybrid materials, LRP can substantially influence the characteristics of intricate biological macromolecules as well as the final products. Therefore, it is vital to choose a method that preserves both the bioactivity of proteins and the properties of the polymers for optimal effectiveness.

Enzyme-Mediated Cross-Linking Enzymes also find application in cross-linking reactions within polymers that creates hydrogel or other types of networks. This method overcomes the drawbacks of chemical cross-linking as it is biocompatible, environmentally friendly and requires mild processing conditions for loading of sensitive biological drugs thus can be used to develop DDS [38].

In general, all the modification techniques as mentioned in the above are very essential in the formation of timely drug delivery system. Certain chemical modifications can require the interaction with the drug and the polymer matrix

thereby controlling the rate of drug release. Physical alterations aid in the improvement of mechanical and surface characteristics of polymers and in ability to interface with biological systems. The enzymatic and the biological composition enable other polymers to biodegrade and allow the delivery of drugs at the specified rates without compromising biocompatibility. Such a synergy of these approaches will allow scientists to obtain polymers that can have desired characteristics appropriate to the corresponding DDS, thus providing better and safer therapeutic outcomes [39].

Physical Modification

Bodies of polymers experience physical modification upon alteration of their morphological features without modification of structural properties. Strategies similar to these are useful in modifying the physical features of polymers and changing the configuration to optimize their functionality in DDS [51].

Surface Modification

They are the processes by which the surface properties of polymers including; wettability, roughness, and biocompatibility can be changed. Other techniques include plasma treatment, corona discharge as well as UV-irradiation which can result to functionalization or altering of the surface energy. Furthermore, the surface-modified polymers are particularly useful in DDS to enhance the adhesion with cells, minimize the protein adsorption or optimize the interaction with tissue [52].

Blending

Polymer blending could be described as the process of combining two or more polymers with a view of producing a material that would possess several characteristics of the said polymers. That, in turn, means that the properties which are mechanical, thermal, or barrier, must not be chemically modified to be improved. In DDS, copolymers are preferred to regulate the rates of drug release as well as the stability of the polymer matrix [53].

Electrospinning

Electrospinning is a novel technique for fabrication of ultrafine polymer fibers with large surface area to volume ratio. The process is done by applying high voltage on a polymer solution and then resulting to emission of fibers. Fibers within DDS by electrospinning can be utilized in scaffold formation for tissue engineering or dressing sites, or drug mat with specific releasing rate [53].

Nanoparticle Formation

Nonetheless, since this is mostly regarded as a method of synthesis, the synthesis of nanoparticles could also be seen to center on physical modification. Macromonomers can be elaborated and solubilized into nanoparticles with the use of evaporation, self-assembly, and nanoprecipitation. These nanoparticles are incorporated within the DDS so as to facilitate the retention of the drug and protect it from the elements while controlling and sustaining a targeted or delayed release [54].

Applications of Modified Polymers

This transformation of polymers is very useful as extends the versatility of polymers and enables their use in areas where specific properties are needed. In this way, modified polymers can meet the high requirements of the various contemporary technologies due to the changes in their chemical composition or the properties of the surface and the physical characteristics. Subsequent sections describe the major uses of modified polymers with a focus on drug delivery systems (DDS).

The implications of polymer modification in drug delivery systems are significant. These modifications enable targeted drug delivery by ensuring medications reach specific sites, enhancing treatment efficacy while minimizing side effects. They also facilitate controlled release, improving patient compliance through sustained drug levels. Additionally, modifications improve biocompatibility, reducing immune responses and prolonging drug circulation. Smart polymers that respond to physiological changes offer precise drug release, while nanoparticle formation enhances drug absorption. Furthermore, biodegradable systems eliminate the need for surgical removal, and modifications in vaccine delivery improve stability and immune responses. Overall, these advancements broaden the therapeutic landscape and enhance patient outcomes.

Drug Delivery Systems (DDS)

The idea of modified polymers has grown into a popular key area of study in the advancement of drug delivery techniques that involve management of the rate of release of a therapeutic agent, targeting, and/or stabilization.

Controlled Release Systems

One of the most significant applications of modified polymers is in controlled release drug delivery systems. By modifying the polymer's chemical structure, researchers can design materials that release drugs over a specific period,

ranging from hours to months. For example, cross-linked hydrogels can swell in response to environmental stimuli, gradually releasing the encapsulated drug. This ensures a consistent therapeutic effect and reduces the frequency of drug administration.

Plasma Modification of Polymer Plasma-treated natural polymer-based drug delivery systems have shown improved biocompatibility, controlled drug release, targeted drug delivery, enhanced cellular uptake, and improved immune response. The modification of naturally derived polymer-based nanoparticles using plasma technology has garnered significant interest recently due to its potential to improve their effectiveness and functionality as drug delivery systems. Polymers like chitosan, alginate, cellulose, and starch are commonly used to create nanoparticles because of their biocompatibility, biodegradability, and minimal toxicity. However, these natural polymer-based nanoparticles often face challenges regarding stability, drug loading capacity, and targeted delivery, which can limit their efficiency as drug carriers. Plasma treatment offers a promising method for enhancing the surface properties of natural polymer-based nanoparticles. This process can introduce functional groups such as amine, carboxyl, and hydroxyl onto the nanoparticle surface, thereby improving their interactions with drugs, cells, and other materials [55]. Additionally, plasma treatment can increase the surface roughness of these nanoparticles, which may enhance their cell adhesion properties and improve drug loading and release profiles. Furthermore, this treatment can bolster the stability of natural polymer-based nanoparticles by enhancing their electrostatic and steric stabilization [56].

Targeted Drug Delivery

Targeted drug delivery systems rely on polymers modified with ligands that recognize and bind to specific receptors on the surface of target cells. These ligands can include antibodies, peptides, or other molecules that guide the drug-loaded polymer to its intended site of action. This approach is particularly beneficial in cancer therapy, where it is essential to deliver toxic drugs directly to tumor cells while minimizing damage to healthy tissues.

Hyperbranched polymers (HBPs) have like dendrimers a three-dimensional globular structure that have attracted the attention from both academia and industry. The advantages of HBPs as compared to linear polymers are their low intrinsic viscosity, low tendency to chain entanglements, smaller hydrodynamic radius, good solubility and high degree of branching (DB) leading to a high number of terminal functional groups. When compared to dendrimers, their structures are irregular with dendritic, linear and terminal units randomly distributed, and their synthesis leads to macromolecules with broad molecular weight distributions. However, HBPs can be easily synthesized in a one-pot reaction and thus are more cost efficient as compared to the multi-step approach for the dendrimers requiring a purification step after each coupling reaction. Furthermore, due to their higher steric hindrance, dendrimers may be more challenging to functionalize than HBPs. HBPs have potential applications in optical, electronic and magnetic, materials, coatings, additives, supramolecular chemistry, and biomedicine. Their features are especially interesting for the development of targeted delivery in the field of drug delivery. The composition of HBPs is tunable at the branching, linear, and terminal units offering a significant degree of freedom in the design of targeted for drug delivery [57].

Biodegradable Implants

Biodegradable polymers, modified to degrade at controlled rates, are used to create drug-loaded implants. These implants can be placed in the body to release drugs locally over an extended period. This approach is useful in treating chronic conditions, where long-term, localized drug delivery is required. The degradation products of these polymers are designed to be non-toxic and are naturally eliminated from the body, reducing the need for surgical removal.

Plasma strategies on biodegradable aliphatic polyesters will be discussed in detail. A large amount of research focuses on one particular plasma-based surface modification technique, namely plasma treatment. Plasma treatments of biodegradable polyesters, the results will be presented of biodegradable polyester individually. The plasma treatments, the other plasma-based modification of polymers [58].

Nanoparticle Carriers

Modified polymers are widely used to fabricate nanoparticles for drug delivery. These nanoparticles can encapsulate drugs, protecting them from degradation and improving their solubility. Surface modifications, such as PEGylation (attachment of polyethylene glycol chains), can enhance the circulation time of nanoparticles in the bloodstream, allowing for more effective drug delivery to target tissues.

The selection of polymer for fabricating nanoparticles depends on the desired size and surface characteristics of the particle as well as the nature of the drug or active ingredient. Physicochemical properties of the polymer determine the fabrication process to be employed [6].

Tissue Engineering

Polymers are employed as a substrate on which cells can grow and form tissues in tissue engineering and have been altered in some manner or the other. Such scaffolds have to

be non-toxic and able to degrade in the organism, as well as to reproduce characteristics of natural tissue ECM.

Tissue engineering approaches that combine biomaterial-based scaffolds with protein delivery systems have provided a potential strategy for improved regeneration of damaged tissue. The success of polymeric scaffolds is determined by the response it elicits from the surrounding biological environment. This response is governed, to a large extent, by the surface properties of the scaffold (Table 1). Surfaces of polymeric scaffolds have a significant effect on protein and cell attachment. Multiple approaches have been developed to provide micrometer to nanometer scale alterations in surface architecture of scaffolds to enable improved protein and cell interactions. Chemical modification of polymeric scaffold surfaces is one of the upcoming approaches that enables enhanced biocompatibility while providing a delivery vehicle for proteins [60].

Scaffold Materials

Polymer can be altered in terms of mechanical strength and elasticity in order to develop scaffold for tissue engineering. These scaffolds offer necessary architecture to cells and afford them the plus factor of getting attached and further developing tissues. Superficial change, including the coating of cell attachment receptor, can also increase cell adhesion and proliferation, these scaffolds are particularly suited for bone, cartilage, and various other tissues regeneration.

Scaffold materials play a crucial role in tissue engineering by providing a three-dimensional framework for cell attachment and growth. Natural polymers like collagen and gelatin enhance biocompatibility and promote cell adhesion, making them ideal for skin and cartilage applications. Synthetic polymers such as PLA and PGA offer tailored mechanical properties and degradation rates. Ceramics, including hydroxyapatite, mimic bone mineral, facilitating bone regeneration. Composite scaffolds combine the benefits of both polymers and ceramics, enhancing strength and bioactivity. Overall, the careful selection of scaffold materials is vital for successful tissue regeneration, with ongoing research focused on innovative designs and fabrication methods [55].

Injectable Hydrogels

Altered polymers are also adapted in the preparation of injectable hydrogels that can simply be applied onto a tissue loss area. These hydrogels can be cross linked in situ to form a gel that can support cell growth and tissue regeneration. That actually these kinds of materials can be injected minimally invasively makes them rather valuable to use in cases when there are some difficulties with accessing the certain area or in those cases when classical surgical operations are impossible to perform.

Medical Devices and Implants

Polymer modified devices and implants stand to benefit from modification since engineers can design these polymers to conform to certain performance standards.

Antimicrobial Coatings

The basis for modification of polymers is to introduce antimicrobial agents into the polymers in order to produce coatings that can inhibit bacterial growth on medical surfaces, devices, or implants. This is very useful in minimizing the

Table 1 Modification of polymer and its application in formulation development

S. No	Polymers	Method	DDS	Reference
1	Chitosan	This system can be created by modifying the liposome's surface with a polymer	Widely used in the form of NPs, hydrogels, and polymeric hydrogel membranes	[59]
2	Alginate	Modulating drug release and imparting thermosensitive properties to liposomes	Used in drug- and protein-delivery systems	[59]
3	Collagen	This system delivers genes by forming stable complexes with either plasmid DNA (pDNA) or small interfering RNA (siRNA)	Used in tissue calcification and as a controlled delivery device for cardiovascular drugs	[59]
4	Cyclodextrin	They are extensively utilized as complexing polymers within liposomes through internal interaction	Particularly in protein and peptide drug delivery and gene delivery	[59]
5	Gelatin	Gelatin complexed with methacrylamide (GelMa) exhibits superior elasticity, promotes cell adhesion, and demonstrates excellent mechanical characteristics	Can be used for targeted drug delivery to cancer sites	[59]
6	Polyacrylamide	Polyacrylamide is commonly utilized as a matrix and carrier within systems that can interact with genetic material through complex formation, particularly in stimuli-responsive applications	It can be administered via injection	[59]

chances of infections connected with devices for example catheters, stents, and orthopedic implants. They can either deliver antimicrobial active agents in a time delayed manner or construct a surface that is unsuitable to support microbial colonization.

The antimicrobial polymers are produced by attaching or inserting an active antimicrobial agent onto polymer backbone via an alkyl or acetyl linker. These are the upcoming new class of disinfectants which can be used even as an alternative to antibiotics in some cases. The treatment of microbial infections becomes more and more difficult because the number of resistant microbial strains as well as that of antibiotic-immune patients grows a lot faster than the number of useable antibiotics. The use of antimicrobial polymers offers promise for enhancing the efficacy of some existing antimicrobial agents and minimizing the environmental problems accompanying conventional antimicrobial agents by reducing the residual toxicity of the agents, increasing their selectivity and prolonging the lifetime of the antimicrobial agents [61].

Biocompatible Coatings

Changed polymers can be used to produce a thin layer that may be dress up for implants including pacemakers, artificial joints, and vascular grafts. These coatings minimize immune responses which are normally detrimental to the body and enhance the integration of implant with the host cells. Other changes to the surface of the device include the addition of anti-inflammatory substances in order to make the devices more biocompatible (Fig. 2).

Drug-Eluting Stents

Drug-eluting stents, which are used for coronary artery disease treatment, are an example of modified polymers at work. Here, the stents are installed with a drug-eluting polymer coating intended to release medication gradually over time to avoid re-narrowing of the arteries (restenosis). Having the polymer in place makes it possible to dispense the drug over time and only, in that specific region of the stent, and hence reduces the off-target side effects of the medication.

Environmental Applications

Similar attributes of the polymers can also be harnessed in the process of environmental preservation and sustainability.

Water Purification

Contaminated water is treated with various water purification systems that utilize polymers whose properties have been adjusted with specific adsorbents which can remove heavy metals, organic pollution, and microorganisms. Such polymers can be functionalized with chelating agents or

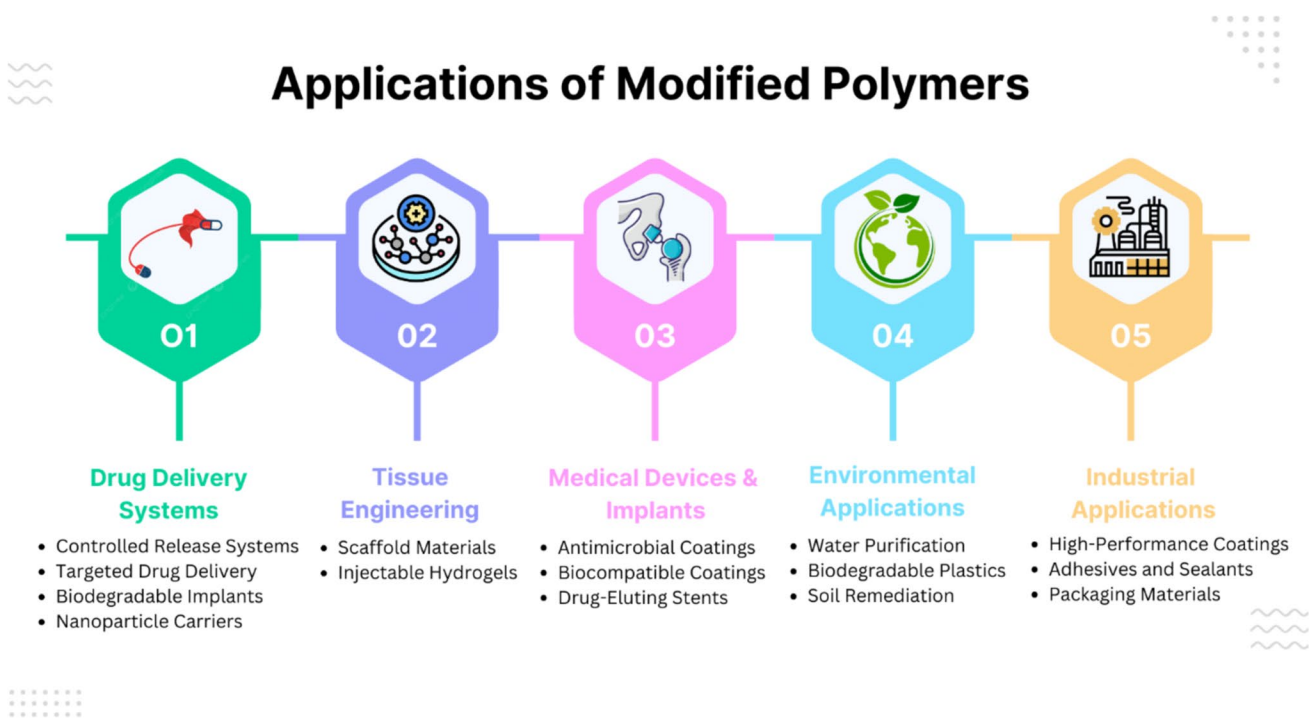

Fig. 2 Applications of modified polymers

antimicrobial agents to provide enhanced water purification. Moreover, the development of bioplastic materials for water treatment technologies is environmentally friendly as it minimizes waste pollution caused by water purification methods.

Biodegradable Plastics

In light of the negative impacts caused by plastic pollution, these modified biodegradable polymers are emerging as potential substitutes for plastics. These polymers incorporate special molecular structures that degrade in the environment faster.

Challenges and Drawbacks

Although there are many advantages to polymer modification, there are also a number of risks and disadvantages that must be addressed. Such problems may negatively affect the practicality, safety, and success of polymer materials with modifications, especially in their use as drug delivery systems.

Production Challenges

Technical considerations however have restricted the general application of polymers in the modification of drug formulations and delivery systems. The modification of polymers is sometimes associated with diverse steps in the synthesis or in the production. Such modification is a form of polymer technology and is taken with consideration of the requisite management of reactions which seems to be a difficult exercise otherwise for large amounts. Reaction conditions for example time and temperature and even the reaction catalysts for the formulation have to be optimized to achieve the desired performance goals. In the same manner, the development of modified polymers may require building up of various intermediates, therefore an increasing risk of cross-contamination and more common inproduct misuse arise. This complexity also features high production costs and low productivity thus the modified polymers are not amenable for mass production.

Environmental and Health Concerns

Polymer modification poses a number of environmental issues, health risks as well as ecological considerations. Some of the chemical modifications tend to use dangerous chemicals or produce hazardous wastes, thus endangering the safety of employees and the environment. For example, volatile organic compounds (VOC) emission comes from some solvents used in the modification of polymer and these are known to cause air pollution and health hazards. On the other hand, waste of modified polymers, especially non-degradable ones, leads to pollution. Such polymers, which may pose health risks in the long terms, did not come on earth surface because they are wait for ailments which people will not fully recover from.

Cost and Scalability Issues

Cost is one of the biggest hurdles in the acceptance of modified polymers by the masses. The presence of complicated processing techniques as well as the need for expensive reagents tend to increase the cost of the modified polymers, which renders it less competitive in relation to other conventional materials. Apart from these reasons, there are some troublesome elements in scaling up the processes of producing modified polymers from the lab level up to the industrial level. The equipment needed for the large-scale production can be expensive, and also relocating the process from a small scale to large scale often requires massive amounts of investment on R&D which focuses on the manufacturing procedures. Such things could impede the enhanced polymers' availability as well as make them expensive especially for the emerging countries.

Stability and Performance Limitations

Stability is a critical feature, which qualifies modified polymers as an important component, especially in the drug delivery devices. It is important that the modified polymers should also be able to stand, temperature, pH, and even humidity changes. On the contrary, certain adaptions can induce some instability thus deterioration and loss of function with time. For example, polymers reinforced with the biodegradable property are favorable for target use, but they degrade in a very short period releasing the drugs within them and therefore not achieving the desired therapeutic effect. Other than these factors, the performance of the modified polymers can be restricted by some other methods.

Regulatory Conditions

Polymer Registration in China

Introducing to manufacture or import any new polymer substance is also submission of registration before the completion of such manufacture or the import proceeds. The particular polymer shall be submitted for registration according to MEE Order No 12 for private businesses. Once a polymer is filed with the polymers registry, it will be delivered and filed focusing on certain elements, if it met the filing parameters and likewise did not fall under the exclusion parameters

based on the five parameters, then polymer record application can be conducted. This applies to polymers containing no more than 2% of monomers or reactants classified as new chemical substances and low-concern polymers. So in case applicants want to apply for polymers that do not fall in this category, they will still have to pose a normal or simplified registration dossier to MEE depending on the amount intended. However if the polymer under consideration however fulfills simultaneous three conditions—these substances does not contain metals except sodium magnesium potassium and calcium does not dissolve in water and certain solvents, is stable at various conditions of pH—toxicology health data ecotoxicology and finally the report on assessment of environmental risk due to the polymer is not needed at the time of normal or simplified registration.

Polymer Registration in Korea

In K-REACH, customers must register the polymer itself, unlike EU REACH. Further, any new polymer substance must be registered before its manufacture or importation. RD Article 8 further stipulates that all existing polymers ≥ 1t/y require pre-registration and registration within specified deadlines. Also, polymers may be excluded from registration to the extent that they are classified as Polymers of Low Concern (PLC); determined on the basado en the structure and composition of the polymer plates. More specifically, exclusions concern polymers with Mn over 10 kDa containing < 5% of oligomers with < 1000 Da < 2% of < 500 Da oligomers and low polymers. Two kinds of exclusion apply predominantly to polymers with Mn of 1000-10-000 where the level of OT eligible components is < 25 and < 10% of < 1000 Da and < 500 Da OT, respectively. Cationic polymers are excluded as are residual polymers with < 10,000 akā rega Mn containing residual hccc & pmc hcc below 0.1%. Polymeric substances that belong to non-risk categories or non-PLCs must be registered specifying to the polymer registrations additional corresponding part with two levels of requirements recession—polymer registrations.

Limitations of Polymer Modifications

Environmental Applications

- *Production Costs:* Advanced polymer modifications can be expensive, making large-scale implementation costly and potentially limiting widespread use.
- *Environmental Impact:* Some modification processes may produce hazardous by-products or require solvents that can be harmful to the environment.
- *Stability Issues:* Modified polymers may have reduced long-term stability, impacting their effectiveness and lifespan in environmental applications.
- *Limited Degradability:* While some modified polymers are designed to be biodegradable, others may persist in the environment, contributing to pollution.

Medical Applications

- *Biocompatibility Concerns*: Modified polymers might trigger immune responses or cause adverse reactions in some patients, affecting their safety and efficacy.
- *Complexity in Manufacturing:* The intricate processes involved in modifying polymers can complicate manufacturing, leading to higher costs and potential scalability issues.
- *Regulatory Challenges:* Obtaining approval for new polymer modifications in medical applications can be lengthy and complex due to stringent regulatory requirements.
- *Variable Performance:* The performance of modified polymers can vary depending on the application, formulation, and patient-specific factors, potentially leading to inconsistent therapeutic outcomes.

Specify trade-offs between improved performance and higher production costs, highlighting how expenses may restrict adoption. Discuss environmental impacts of by-products from modification processes and the long-term stability of modified polymers. In medical contexts, address variability in patient responses and its implications for therapeutic outcomes. Lastly, emphasize the need for extensive research to establish standardized protocols and guidelines for polymer modifications, underscoring the importance of further investigation in this evolving field.

Future Prospects and Innovations

Despite the challenges associated with polymer modification, ongoing research and technological advancements offer promising prospects for the future. Innovations in drug delivery systems, emerging modification techniques, and the development of sustainable polymers are poised to address current limitations and expand the applications of modified polymers.

Innovations in Systems for Drug Delivery

The evolution of drug delivery systems is highly depended upon the evolution of polymer modifications. New drugs with better delivery outcomes are engineered through sophisticated design of polymers. For instance, the introduction of temperature and pH-sensitive polymers is a welcome

progress since it provides an opportunity to control the rate and targeted release of drugs from the carrier. Furthermore, new approaches in nanotechnology are making it possible to generate polymeric nanoparticles that can be targeted to any cell and tissue with high precision. These innovations can help enhance the safety and effectiveness of drug delivery systems, hence improving the recovery of the patients.

New Strategies for the Modification of Polymers

In polymer modification, many newcomers have emerged and are helping in lengthening the chances of polymer's structure and usage. One suitable procedure is the application of click chemistry for building complex polymers, which has proved to be very efficient and specific. Click chemistry ensures that certain functional groups can be attached to polymers in a precise manner so that specific materials with desired functions can be synthesized. 3D printing is yet another interesting technique which revolves around the construction of particular polymer structural components.

Sustainable and Eco-Friendly Polymers

According to most polymers research today is sustainability and there is high emphasis on building alternative bio based polymers. Scientists are striving to develop polymers that are not only compostable but also sourced from renewable resources such as biomass. These beneficial polymers can help in minimizing the depletion of conventional plastic materials and help toward the circular economy indeed. Also, the development of novel technologies for recycling polymers makes it possible to extract and use them again, even after reheating a polymer with different additives to create films, tubes, etc. With the rising awareness of sustainable materials, the market for such eco-friendly materials is highly expected to grow along with their development and adoption, providing a better and more reliable solution to a wide range of applications.

To further the advancement of polymer modifications and improve drug delivery systems, future research should concentrate on several important areas. First, it is crucial to investigate the creation of multifunctional polymers capable of meeting various therapeutic requirements simultaneously. Additionally, exploring hybrid materials that merge the characteristics of both natural and synthetic polymers could lead to innovative solutions with enhanced biocompatibility and functionality. There is also a need for more research into the long-term stability and degradation profiles of newly developed biodegradable polymers to ensure their efficacy in clinical settings. Moreover, examining the integration of intelligent technologies, such as responsive drug delivery systems that can react to physiological changes in real-time, is essential. Finally, fostering stronger collaborations between academic institutions and industry partners will facilitate the practical application of research findings, ensuring that advancements in polymer modifications lead to meaningful improvements in patient care.

Conclusion

Polymer modification is an effective strategy for developing materials with tailored properties for various applications, particularly in drug delivery systems. This paper emphasizes the benefits of these modifications, including enhanced targeted delivery and controlled release, which significantly improve therapeutic outcomes. However, challenges such as production complexity, environmental impact, cost, and stability remain. Ongoing research is actively addressing these issues by refining drug delivery methods, exploring innovative modification techniques, and developing sustainable polymers. The future of polymer modification appears promising, as advancements aim to overcome current limitations and expand applications. This study demonstrates that specific polymer modifications enhance performance and safety, reinforcing the significance of this field for both healthcare advancements and environmental solutions.

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Data Availability The datasets supporting this study are derived from publicly available scholarly articles indexed in Scopus and Web of Science.

Declarations

Conflict of interest The authors declare that there is no conflict of interest.

Human and Animal Rights Not Applicable, no experiments involving human participants or animals were conducted.

Informed Consent Not Applicable.

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